

PARAG GHOSH

+91-9366643548 | ghosh.parag2025@gmail.com | linkedin.com/in/parag-ghosh | paragghosh99.github.io

TECHNICAL SKILLS

Languages: Python (OOP, data structures, complexity analysis), Java, C++, SQL
Backend and APIs: FastAPI, REST APIs, Authentication Systems, Service Architecture
Databases: MySQL, SQL Server (Indexing, Query Optimization), Firestore
Systems: Linux, Distributed Systems, Fault Tolerance, Monitoring
DevOps: Docker, CI/CD, GitHub/GitLab, GitHub Actions, Bash, Cloud Run
Core CS Fundamentals: Data Structures (**Trees, Graphs, Heaps**), Algorithms (**DFS, BFS, DP, Sliding Window, Recursion, Backtracking**), Time & Space Complexity

EXPERIENCE

- Ericsson India Global Services Pvt. Ltd** — *Developer* Oct 2023 - Feb 2025
- Designed and owned a concurrent configuration metadata ingestion service for global infrastructure, enforcing schema contracts and validation rules to gatekeep inputs to a large-scale energy-optimization engine.
 - Built a multi-stage validation pipeline with secure schema enforcement to reject malformed configs at ingestion, reducing invalid inputs by ~40% and eliminating a major failure source.
 - Drove schema evolution across interlinked datasets with backward-compatible design, enabling scalable onboarding and cutting integration turnaround from 3–5 days to ~1–2 days.
 - Hardened upstream data integrity via validation gates and fault-tolerant design, reducing pipeline failure rates by ~65% and improving system reliability.
- Ericsson India Global Services Pvt. Ltd** — *Associate Engineer Trainee* Aug 2022 - Sep 2023
- Designed and implemented a high-throughput, concurrent batch processing system for 4G/5G network data, powering automated infrastructure scaling and optimization decisions.
 - Built optimized SQL pipelines (joins, indexing, stored procedures) processing 100K+ records across 8+ tables, generating reports consumed by downstream systems.
 - Resolved database contention and table locks using multithreaded query execution and staged queries, reducing batch runtime from ~45 to ~30 minutes (35%).
 - Improved execution reliability by ~30% using fault-tolerant processing (retry logic, transactions), reducing manual intervention from daily to weekly.
- Ericsson India Global Services Pvt. Ltd** — *Graduate Engineer Trainee Intern* Jun 2022 - Aug 2022
- Built a Python-based data pipeline restructuring complex joins into staged, validated SQL queries, achieving 3× faster execution.

PROJECTS

- Code MRI – Repository Analysis Engine** | *FastAPI, Cloud Run, Vertex AI* | *Devpost Gemini Live Agent Hackathon Participant*
- Engineered an architecture scanner that crawls 500+ repo files into JSON + dependency graphs, reducing latency 65% (~60s → less than 20s) by replacing browser-agent screenshots with Git tree parsing.
 - Built a NetworkX-based graph analysis layer to compute module centrality and coupling metrics, surfacing single points of failure and database bottlenecks through deterministic analysis.
 - Built a scalable cloud-native workflow with caching and async/concurrent execution to maintain API responsiveness, with a React frontend for insights.
- Scalable Task & Identity Platform** | *FastAPI, SQLAlchemy, JWT, Docker, CI/CD*
- Built GitHub Actions CI/CD pipeline with PR/issue-based workflows to automate build, test, and validation, ensuring reliable releases.
 - Designed a layered Routes → Services → Repositories architecture with SQLAlchemy and Pydantic, enforcing separation of concerns, schema validation, and maintainable service boundaries.
 - Built GitHub Actions CI/CD pipeline to automate build, test, and validation workflows, ensuring reliable releases.

EDUCATION

Sikkim Manipal Institute of Technology
B.Tech in Computer Science — CGPA: 7.6

Sikkim, India
Aug 2018 - May 2022